

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T52 · Transformations

With Answers

Questions with full worked solutions

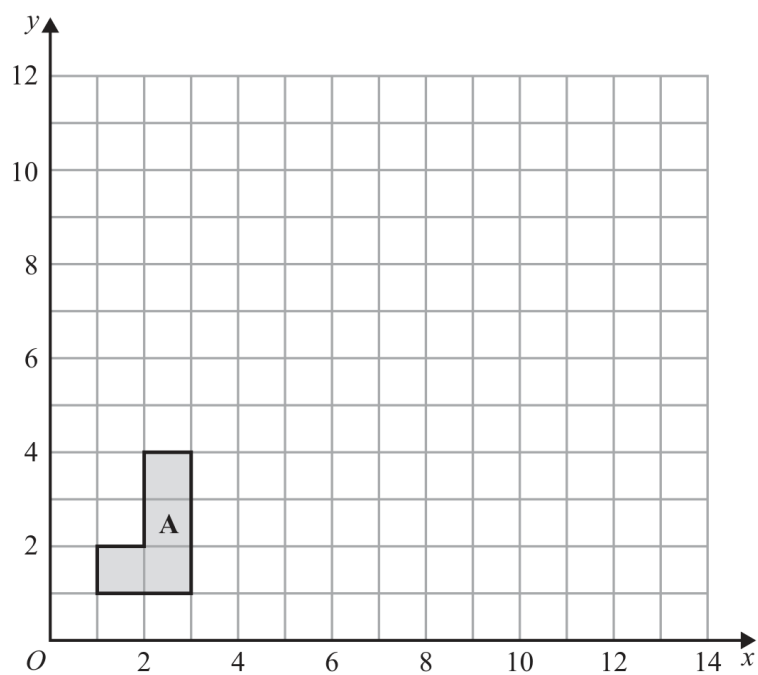
■ 17 pages

FAMILY MAP

Transformations

F01	Construct an enlargement	2 Q 4 marks
F02	Describe fully one transformation	3 Q 11 marks
F03	Construct an isometry on a coordinate grid	1 Q 2 marks
F04	Recover coordinates or parameters from a known transformation	1 Q 3 marks

2



(a) On the grid, enlarge shape A with scale factor 3 and centre $(0, 1)$

(2)

Worked Solution 17

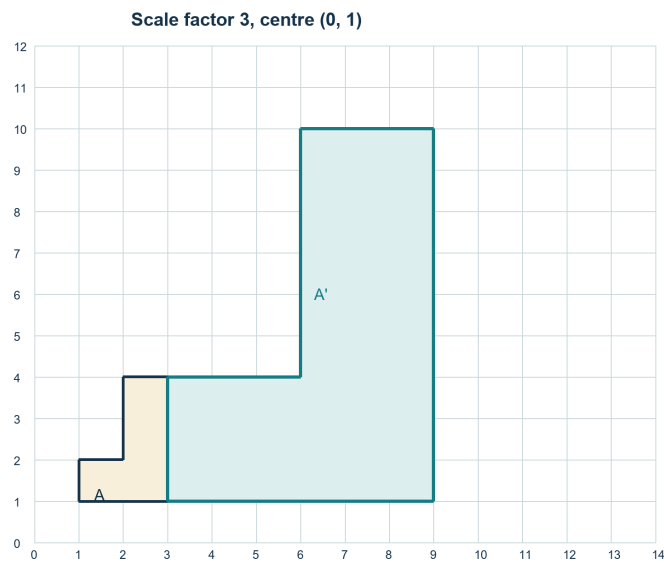
Enlargement from a centre

UNIT 2
2 marks**Apply the map**

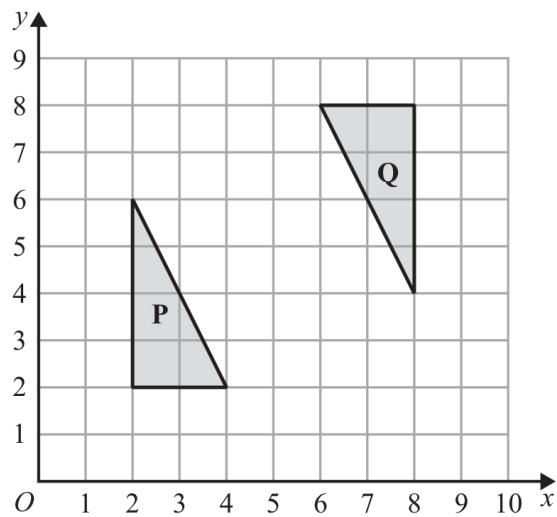
$$(x, y) \mapsto (3x, 3y - 2) \text{ for scale factor 3 about } (0, 1).$$

Image vertices

$$(3, 1), (9, 1), (9, 10), (6, 10), (6, 4), (3, 4).$$

**Final answer**

The enlarged shape has vertices $(3, 1), (9, 1), (9, 10), (6, 10), (6, 4), (3, 4)$



(b) Describe fully the single transformation that maps triangle **P** onto triangle **Q**

(3)

Worked Solution 18

Rotation of a triangle

UNIT 2**3 marks****Identify the transformation** $P \rightarrow Q$ is a 180° rotation about $(5, 5)$.**Check one vertex**

$$(2, 2) \mapsto (2 \cdot 5 - 2, 2 \cdot 5 - 2) = (8, 8), \text{ matching } Q.$$

Final answerRotation 180° about $(5, 5)$

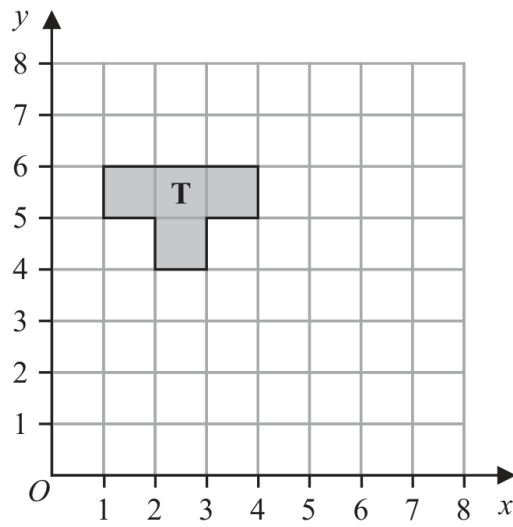
Question 23

4MA1-2H Q3 (a)

ORIGINAL QUESTION UNIT 2

2 marks

3

(a) Reflect shape **T** in the line $y = x$

(2)

Worked solution 23Reflection in $y = x$

MODULAR UNIT 2

2 marks

(a) – Reflection in $y = x$

Apply the reflection rule

Reflection in $y = x$ maps $(x, y) \mapsto (y, x)$.

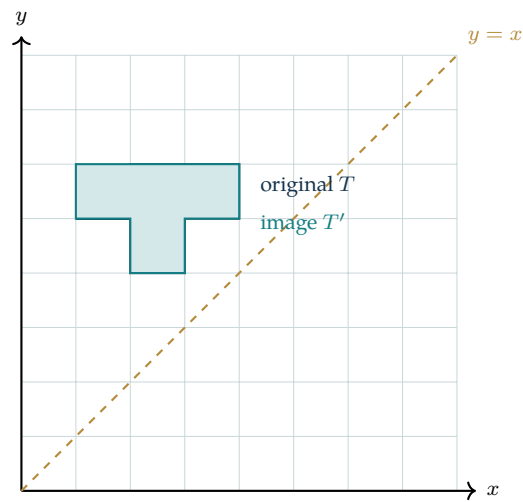
Plot the image

The image vertices are $(4, 2), (4, 3), (5, 3), (5, 4), (6, 4), (6, 1), (5, 1), (5, 2)$; join them in order.**Final answer**Reflected shape with vertices $(4, 2), (4, 3), (5, 3), (5, 4), (6, 4), (6, 1), (5, 1), (5, 2)$

Completed diagram 23Reflection in $y = x$

MODULAR UNIT 2

2 marks

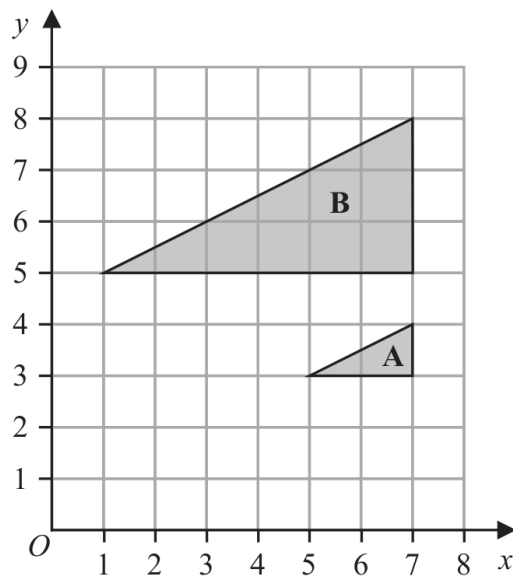
Completed solution diagram: Reflection in $y = x$ 

Question 24

4MA1-2H Q3 (b)

ORIGINAL QUESTION UNIT 2

3 marks



(b) Describe fully the single transformation that maps triangle A onto triangle B

(3)

Worked solution 24

Describe the enlargement

MODULAR UNIT 2

3 marks

(b) – Describe the enlargement

Compare corresponding lengths

Every side of B is three times the corresponding side of A, with the same orientation.

Locate the centre

The centre is $(7, 2)$; for example $(7, 3)$ maps to $(7, 5)$ with scale factor 3.

Final answer

Enlargement, scale factor 3, centre $(7, 2)$

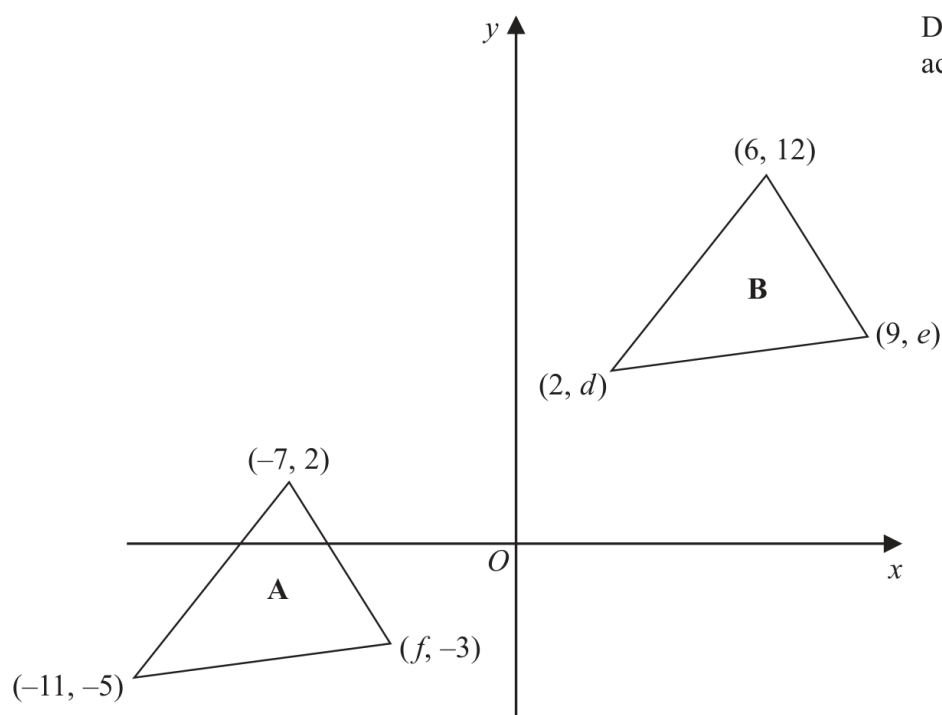
Question 04

4MA1-1H Q4

ORIGINAL QUESTION

3 marks

- 4 The diagram shows a sketch of triangle **A** and triangle **B** on a coordinate grid.



- (a) Given that triangle **A** has been translated to give triangle **B**, work out the value of d , the value of e and the value of f

 $d = \dots\dots\dots$ $e = \dots\dots\dots$ $f = \dots\dots\dots$

(3)

Worked Solution 04

Translation coordinates

MODULAR UNIT 2

3 marks

Read the translation

$$(6, 12) - (-7, 2) = (13, 10).$$

Apply it to the other vertices

$$d = -5 + 10 = 5, \quad e = -3 + 10 = 7, \quad f = 9 - 13 = -4.$$

Final answer

$$d = 5, \quad e = 7, \quad f = -4$$

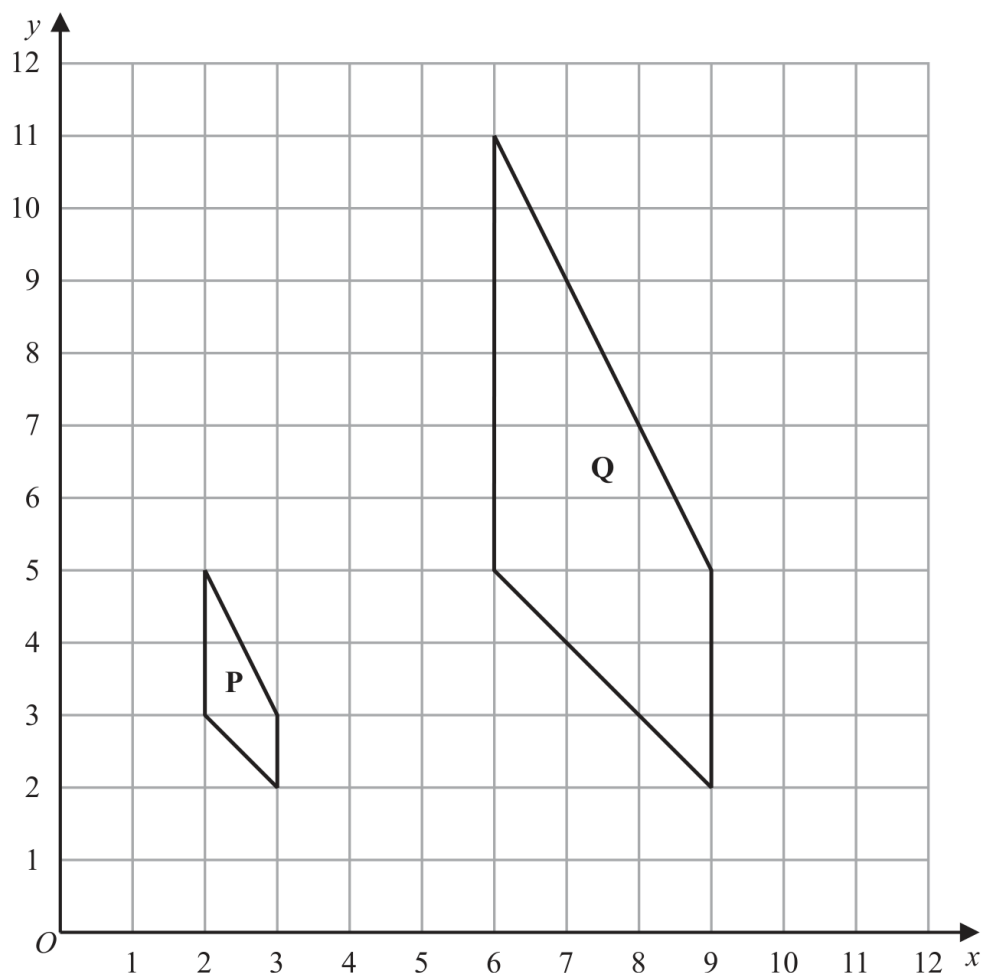
Question 05

4MA1-1H Q4

ORIGINAL QUESTION

5 marks

The diagram shows shape **P** and shape **Q** drawn on a grid.



(b) Describe fully the single transformation that maps shape **P** onto shape **Q**

(3)

(c) On the grid above, rotate shape **P** 90° clockwise about (3, 5)
Label your shape **R**

(2)

Worked Solution 05

Enlargement and rotation

MODULAR UNIT 2

5 marks

Describe P to QEnlargement, scale factor 3, centre $(0, 2)$.**Rotate P**

$$(x, y) \mapsto (3 + (y - 5), 5 - (x - 3)).$$

Plot the image

$$R : (0, 5), (1, 6), (3, 6), (1, 5).$$

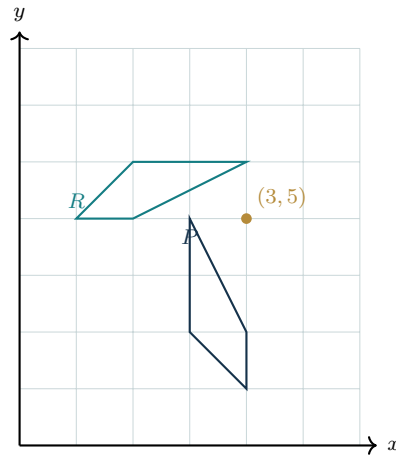
Final answerEnlargement $(k = 3, C = (0, 2))$; $R : (0, 5), (1, 6), (3, 6), (1, 5)$

Completed diagram 05

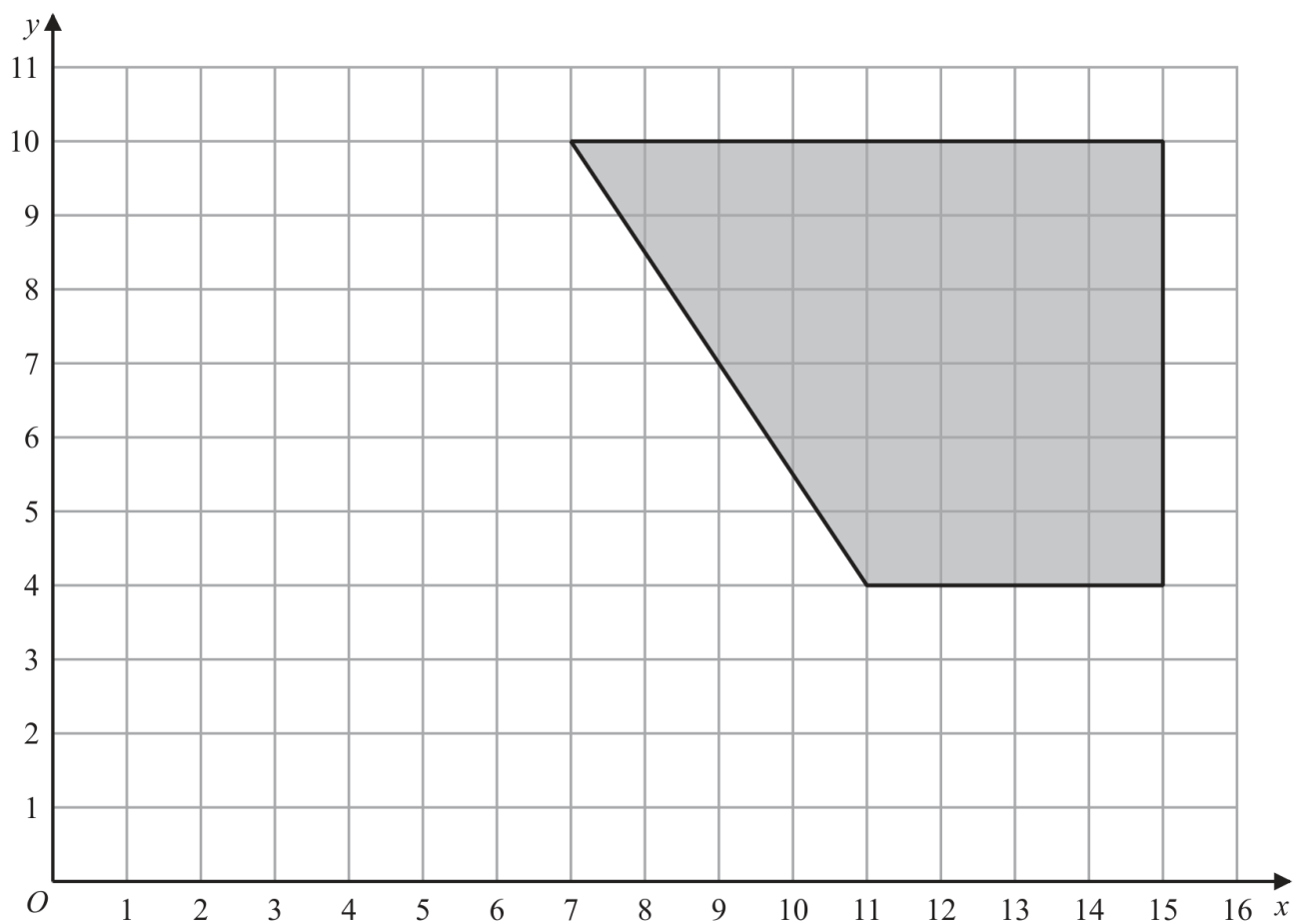
Enlargement and rotation

MODULAR UNIT 2

5 marks

Completed solution diagram: Rotation image on the coordinate grid

8



On the grid, enlarge the shaded shape with scale factor $\frac{1}{2}$ and centre (1, 2)

Worked solution

January 2020 | Question 8 | 2 marks

Family: Construct an enlargement • Variant: Positive fractional scale factor

Method / working

1. Use $\text{centre} + \frac{1}{2}(\text{point} - \text{centre})$ for each vertex.
2. The image vertices are $(4,6), (8,6), (8,3), (6,3)$.

Final answer: $(4,6), (8,6), (8,3), (6,3)$